Near-Tight Guarantees for Sparse Matrices in Streaming PCA

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1 Overview

Principal Component Analysis is the problem of finding the k-dimensional subspace of the largest variance in a dataset. For large datasets, it is impractical to store the entire matrix $X \in \mathbb{R}^{n \times d}$, or even the covariance matrix $\frac{1}{n} X^\top X \in \mathbb{R}^{d \times d}$. Our main objective is to analyze algorithms for PCA using only $\tilde{O}(d)$ space in the streaming setting, where data points arrive sequentially.

Oja's algorithm begins with a random unit vector v_0, iteratively updating with the rule $v_i = v_{i-1} + \eta x_i x_i^\top v_{i-1}$, with the hope that by the end of the algorithm, $\frac{v_n}{||v_n||} \approx v_*$, our top eigenvector. While Oja's algorithm performs well in practice, it has only been extensively studied in the stochastic setting, where data points are drawn i.i.d from a nice distribution.

In a novel analysis, Price & Xun (FOCS 2024) bounded the performance of Oja's algorithm in the arbitrary data setting. So long as the spectral ratio $R = \lambda_1/\lambda_2$ is sufficiently large, Oja's algorithm can converge to the top principal component. Their main result can be summarized as follows:

Theorem 1.1 *(Theorem 1.2 of [1]) For a matrix $X \in \mathbb{R}^{n \times d}$ with spectral ratio satisfying $R = \lambda_1/\lambda_2 = \Omega(\log n \log d)$, there exists an algorithm using $\tilde{O}(d)$ space obtaining an approximation of the top principal component with error $o_d(1)$.*

The analysis of this upper bound has been marginally improved by Kacham & Woodruff [2], who use a modified version of Oja's algorithm combined with row-norm sketching to get $n = \mathrm{poly}(d)$ in expectation, which yields spectral ratio requirement $R = \Omega(\log^2 d)$. This analysis also provides the best-known lower bound of $O(\log d / \log \log d)$.

In this work, we improve the analysis of Price & Xun for matrices with a "sparse representation", i.e. matrices with $O(1)$ nonzero entries per row under some orthogonal basis, showing Price & Xun's modified version of Oja's algorithm works on such inputs so long as $R = \Omega(\log d)$. We will follow the notational conventions and assumptions of [1].

2 Main Results

We first provide an extremely brief summary of Price & Xun's analysis. We begin with the main technical finding:

Lemma 2.1 *(Lemma 2.5 of [1]) If $v_0 = v_*$, then*

$$\eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1} \rangle \lesssim \sigma_2^2 \log^2 n \log ||v_n||$$

As per Lemma 2.6, one gets that

$$\log ||v_n||^2 \geq \frac{1}{4}\sigma_1 - \eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1} \rangle^2 \geq \frac{1}{4}\sigma_1 - \sigma_2^2 \log^2 n \log ||v_n|| \tag{1}$$

$$\therefore \ \log ||v_n|| \gtrsim \frac{\sigma_1}{1 + \sigma_2^2 \log^2 n}$$

Finally, Claim 3.3 and Lemma 2.1 allow us to obtain

$$||P\hat{v}_n|| \leq \sqrt{\sigma_2} + \frac{||Pv_0||}{||v_n||}$$

The assumption that $v_0 = v_*$ leverages the linearity of Oja's algorithm, allowing us to hone in on the v_* component of our initial random vector. The downside is that since the expected magnitude of this component is $O(1/\sqrt{d})$, we must make up for an additional poly(d) factor of growth. But for our vector to grow polynomially in d, we must have that $\log ||v_n|| \simeq \sigma_1$, as $\sigma_1 = \Omega(\log d)$ by assumption. In equation (1), this requires $\sigma_2 = O(1/\log n)$, which gives us the $\log n$ term in $R = \Omega(\log n \log d)$.

We observe that $\eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1} \rangle^2$ cannot be large without $\hat{v}_{i-1}$ rotating into many distinct directions, as variance in any fixed direction is bounded by σ_2. Indeed, the proof of Lemma 2.5 in [1] heavily relies on Lemma 2.2, which shows that $\hat{v}_{i-1}$ cannot rotate much without increasing its norm. We double down on this intuition by noting that for sparse matrices, the shifts in v_{i-1} occurring at each iteration are not particularly complex, so the relationship between growth and rotation can be made much more explicit.

In Theorem 2.2, we show that when each data point can be represented as a linear combination between v_* and exactly 1 orthogonal direction, we can improve the analysis to allow for $R = \Omega(\log d)$. We then generalize this in Theorem 2.3 to streams where each data point is a linear combination of v_* and $O(1)$ orthogonal directions. Note this is synonymous with the stream having a sparse representation under the basis consisting of its right singular vectors.

2.1 Bound for 1 dimension

Theorem 2.2 . *Given data points $x_1, ..., x_n$ with top principal component v_* and remaining principal components $w_1, w_2, ..., w_{d-1}$, where each x_i satisfies $|\{j : \langle x_i, w_j \rangle \neq 0\}| = 1$ for all*

$i \in [n]$, we have that

$$\eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1}\rangle^2 \lesssim \sigma_2^2 \log ||v_n||$$

Proof. Let $h(i) : [n] \to [d-1]$ map each x_i to its orthogonal direction; that is, $\langle x_i, w_j \rangle \neq 0 \iff j = h(i)$. Denote $S_j = \{i : \langle x_i, w_j\rangle \neq 0\}$ be the set of all indices contributing to the jth orthogonal direction, for all $j \in [d-1]$. With this notation and since each x_i only contributes to 1 orthogonal direction, we can rewrite $\langle x_i, P\hat{v}_{i-1}\rangle^2 = \langle x_i, w_{h(i)}\rangle^2 \langle \hat{v}_{i-1}, w_{h(i)}\rangle^2$. Hence, we have that

$$\eta \sum_{i=1}^{n} \langle x_i, w_{h(i)}\rangle^2 \langle \hat{v}_{i-1}, w_{h(i)}\rangle^2 \simeq \eta \sum_{i=1}^{n} \langle x_i, w_{h(i)}\rangle^2 \frac{\left(\eta \sum_{\substack{x_k \in S_{h(i)} \\ k<i}} \langle x_k, v_{k-1}\rangle \langle x_k, w_{h(i)}\rangle\right)^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2}$$

Here, we use that $||v_{i-1}||^2 = 1 + \frac{1}{z}\eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2$ for some constant $z \in [1, 3]$. Then, applying Cauchy-Schwarz on the numerator, we get

$$\leq \eta \sum_{i=1}^{n} \langle x_i, w_{h(i)}\rangle^2 \eta \sum_{\substack{k<i \\ x_k \in S_{h(i)}}} \langle x_k, w_{h(i)}\rangle^2 \frac{\eta \sum_{\substack{k<i \\ x_k \in S_{h(i)}}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2}$$

Observing the sum of $\eta \langle x_k, w_{h(i)}\rangle^2$ is at most σ_2, we can move it outside the summation with the upper bound

$$\leq \sigma_2 \eta \sum_{i=1}^{n} \langle x_i, w_{h(i)}\rangle^2 \frac{\eta \sum_{\substack{k<i \\ x_k \in S_{h(i)}}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2}$$

Breaking this summation to be across the $d-1$ orthogonal directions rather than on the data point indices, we get

$$= \sigma_2 \eta \sum_{j=1}^{d-1} \sum_{x_i \in S_j} \langle x_i, w_j\rangle^2 \frac{\eta \sum_{\substack{k<i \\ x_k \in S_j}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2}$$

Note the inner fraction is the ratio of growth from direction j with respect to the total growth up until index i. Now, let $i_*^{(j)}$ denote the arg max index across i of the growth ratio for fixed direction j, so our summation is bounded above by

$$\leq \sigma_2 \eta \sum_{j=1}^{d-1} \sum_{x_i \in S_j} \langle x_i, w_j\rangle^2 \frac{\eta \sum_{\substack{k<i_*^{(j)} \\ x_k \in S_j}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1}\rangle^2}$$

Since the optimal growth ratio is only dependent on j and not i, and since $\sum_{x_i \in S_j} \eta \langle x_i, w_j\rangle^2 \leq \sigma_2$, we get the upper bound

$$\leq \sigma_2^2 \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{k<i_*^{(j)} \\ x_k \in S_j}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1}\rangle^2}$$

It remains to be shown that the summation is $O(\log ||v_n||)$. WLOG, label our directions in the order of the $i_*^{(j)}$'s, so $i_*^{(1)} < ... < i_*^{(d-1)}$. Let $c_j = \frac{\eta \sum_{\substack{k<i_*^{(j)} \\ x_k \in S_j}} \langle x_k, v_{k-1}\rangle^2}{z+\eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1}\rangle^2}$ be the growth ratio for a given direction j. We have that

$$c_{d-1} = \frac{\eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \in S_{d-1}}} \langle x_k, v_{k-1}\rangle^2}{z + \eta \sum_{k<i_*^{(d-1)}} \langle x_k, v_{k-1}\rangle^2}$$

$$\therefore \eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \in S_{d-1}}} \langle x_k, v_{k-1}\rangle^2 = c_{d-1} \left(z + \eta \sum_{k<i_*^{(d-1)}} \langle x_k, v_{k-1}\rangle^2 \right)$$

Removing the terms of the LHS from the right and dividing the coefficient, we get

$$\eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \in S_{d-1}}} \langle x_k, v_{k-1}\rangle^2 = \frac{c_{d-1}}{1-c_{d-1}} \left(z + \eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \notin S_{d-1}}} \langle x_k, v_{k-1}\rangle^2 \right)$$

Adding $z + \eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \notin S_{d-1}}} \langle x_k, v_{k-1}\rangle^2$ on both sides, we get

$$z + \eta \sum_{k<i_*^{(d-1)}} \langle x_k, v_{k-1}\rangle^2 = \frac{1}{1-c_{d-1}} \left(z + \eta \sum_{\substack{k<i_*^{(d-1)} \\ x_k \notin S_{d-1}}} \langle x_k, v_{k-1}\rangle^2 \right)$$

which, by our ordering of indices, is at least

$$\geq \frac{1}{1-c_{d-1}} \left(z + \eta \sum_{\substack{k<i_*^{(d-2)} \\ x_k \notin S_{d-1}}} \langle x_k, v_{k-1}\rangle^2 \right)$$

$$\geq \frac{1}{1-c_{d-1}} \left(\eta \sum_{\substack{k<i_*^{(d-2)} \\ x_k \notin S_{\geq d-2}}} \langle x_k, v_{k-1}\rangle^2 \right)$$

(Here, we denote $S_{\geq m} = \bigcup_{j=m}^{d-1} S_j$). Which we can recursively expand upon to get

$$\geq \left(\prod_{j=2}^{d-1} \frac{1}{1-c_j} \right) \left(z + \eta \sum_{\substack{k<i_*^{(1)} \\ x_k \in S_1}} \langle x_k, v_{k-1}\rangle^2 \right)$$

Using the bound $\frac{1}{1-x} > e^x$ for $x \in [0,1]$, we get

$$\geq e^{c_2+\ldots+c_{d-1}} z \geq \frac{1}{e^{c_1}} e^{c_1+\ldots+c_{d-1}} z \geq \frac{z}{e} e^{c_1+\ldots+c_{d-1}}$$

Now, since $||v_n||^2 \simeq z + \eta \sum_{k\leq n} \langle x_k, v_{i-1}\rangle^2 \geq \eta \sum_{k<i_*^{(d-1)}} \langle x_k, v_{k-1}\rangle^2$, we have that

$$\log ||v_n|| \gtrsim e^{c_1+\ldots+c_{d-1}}$$

$$\therefore \frac{1}{\sigma_2^2} \eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1}\rangle^2 \leq \sum_{j=1}^{d-1} c_j \lesssim \log ||v_n||$$

as desired. □

Observe that the analysis doesn't use that $w_{h(i)}$ are the principal components of X. In fact, we could pick any arbitrary basis of $(v_*)^\perp$ to no loss, i.e. the theorem holds for matrices that are sparse under *any* orthogonal basis. Additionally, note that this analysis is made algebraically possible due to an implicit Cauchy-Schwarz on $\langle x_i, P\hat{v}_{i-1}\rangle^2 = \left(\langle x_i, w_{h(i)}\rangle\langle \hat{v}_{i-1}, w_{h(i)}\rangle\right)^2$, which shouldn't hurt us asymptotically so long as each x_i contributes to only $O(1)$ directions.

As per equation 1, this theorem ultimately shows that $R = \Omega(\log d)$ is sufficient. Using this as a baseline framework, we now generalize this approach.

2.2 Generalizing to $O(1)$ dimensions & beyond

Theorem 2.3 *Given data stream $x_1, ..., x_n$ with principal component v_* and arbitrary basis $w_1, w_2, ..., w_{d-1}$ spanning $(v_*)^\perp$, where each x_i satisfies $|\{j : \langle x_i, w_j\rangle \neq 0\}| \leq T$, we have that*

$$\eta \sum_{i=1}^{n} \langle x_i, P\hat{v}_{i-1}\rangle^2 \lesssim \sigma_2^2 T \log ||v_n||$$

Proof. Observe that

$$\eta \sum_{i=1}^{n} \left(\sum_{j=1}^{d-1} \langle x_i, w_j\rangle\langle \hat{v}_{i-1}, w_j\rangle \right)^2 = \eta \sum_{i=1}^{n} \left(\sum_{j\in h(i)} \langle x_i, w_j\rangle\langle \hat{v}_{i-1}, w_j\rangle \right)^2$$

We know $|h(i)| \leq T$, so by Cauchy-Schwarz on the inner summation, we proceed as in Theorem 2.2 to get:

$$\eta T \sum_{i=1}^{n} \sum_{j\in h(i)} \langle x_i, w_j\rangle^2 \langle \hat{v}_{i-1}, w_j\rangle^2$$

$$\simeq \eta T \sum_{i=1}^{n} \sum_{j\in h(i)} \langle x_i, w_j\rangle^2 \frac{\left(\eta \sum_{\substack{x_k\in S_j \\ k<i}} \langle x_k, v_{k-1}\rangle\langle x_k, w_j\rangle\right)^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1}\rangle^2}$$

$$
\begin{aligned}
&\leq \eta T \sum_{i=1}^{n} \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \eta \sum_{\substack{x_k \in S_j \\ k<i}} \langle x_k, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1} \rangle^2} && \text{(Cauchy)} \\
&\leq \eta \sigma_2 T \sum_{i=1}^{n} \sum_{j \in h(i)} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1} \rangle^2} \\
&= \eta \sigma_2 T \sum_{j=1}^{d-1} \sum_{i=1}^{n} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i} \langle x_k, v_{k-1} \rangle^2} \\
&\leq \eta \sigma_2 T \sum_{j=1}^{d-1} \sum_{i=1}^{n} \langle x_i, w_j \rangle^2 \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i_*^{(j)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1} \rangle^2} \\
&\leq \eta \sigma_2 T \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i_*^{(j)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1} \rangle^2} \sum_{i=1}^{n} \langle x_i, w_j \rangle^2 \\
&\leq \sigma_2^2 T \sum_{j=1}^{d-1} \frac{\eta \sum_{\substack{x_k \in S_j \\ k<i_*^{(j)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j)}} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}
$$

To evaluate the final sum, we fix an ordering of each $h(i)$ and define $S_{j,p}$ as the set of all x_k such that x_k appears as the pth index in $h(i)$ (for $1 \leq p \leq T$), and $\langle x_k, w_j \rangle \neq 0$. We then define $i_*^{(j,p)}$ as the index obtaining the maximum growth ratio across $S_{j,p}$. Then, observing that $i_*^{(j)} = i_*^{(j,p)}$ for some p, we can upper bound the sum as

$$
\begin{aligned}
&\sigma_2^2 T \sum_{j=1}^{d} \sum_{p=1}^{T} \frac{\eta \sum_{\substack{x_k \in S_{j,p} \\ k<i_*^{(j,p)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j,p)}} \langle x_k, v_{k-1} \rangle^2} \\
&= \sigma_2^2 T \sum_{p=1}^{T} \sum_{j=1}^{d} \frac{\eta \sum_{\substack{x_k \in S_{j,p} \\ k<i_*^{(j,p)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j,p)}} \langle x_k, v_{k-1} \rangle^2}
\end{aligned}
$$

Defining $c_{j,p} = \frac{\eta \sum_{\substack{x_k \in S_{j,p} \\ k<i_*^{(j,p)}}} \langle x_k, v_{k-1} \rangle^2}{z + \eta \sum_{k<i_*^{(j,p)}} \langle x_k, v_{k-1} \rangle^2}$ and invoking Theorem 2.2 on the inner sum, we get

$$
\begin{aligned}
\sigma_2^2 T \sum_{p=1}^{T} \sum_{j=1}^{d} c_{j,p} \leq \sigma_2^2 T \sum_{p=1}^{T} \log ||v_n|| \simeq \sigma_2^2 T \log ||v_n|| \\
\simeq \sigma_2^2 T^2 \log ||v_n||
\end{aligned}
$$

as desired. □

Setting $T = O(1)$ in Theorem 2.3 gives us our intended result for sparse matrices.

References

[1] Eric Price, Zhiyang Xun. *Spectral Guarantees for Adversarial Streaming PCA*. FOCS. 2024. URL: https://arxiv.org/abs/2408.10332.

[2] Praneeth Kacham, David P. Woodruff. *Approximating the Top Eigenvector in Random Order Streams*. NeurIPS. 2024. URL: https://arxiv.org/abs/2412.11963v1.